Weihan Li

weihanli@gatech.edu | weihanlikk.github.io | Google Scholar | github.com/WeihanLikk

Research Summary

I work on **robot learning for agentic, dexterous control**: learning contact-rich skills for 20-DoF multi-fingered hands from human motion capture with large-scale reinforcement learning in GPU simulation, transferring them to real hardware, and building them as a skill layer that a high-level VLM or VLA agent can call. Before robotics, I worked on **machine learning for neural and behavioral data**: multimodal motion encoders and foundation models for animal behavior from pose and video, and scalable latent dynamical models of multi-region neural activity, with 11 publications at NeurIPS, ICML, and ICLR (4 first-author, including an ICML 2025 Oral).

Simulation demos: weihanlikk.github.io/demos

Education

Georgia Institute of Technology, Ph.D. in Machine Learning — Sept 2023 – Present

- Advisor: Prof. Anqi Wu. M.S. in Computer Science, 2025.

Zhejiang University, Research Assistant — Sept 2020 – May 2023

- Advisor: Prof. Gang Pan. Neural decoding and brain–computer interfaces.

Zhejiang University, B.Eng. in Computer Science and Technology — Sept 2016 – June 2020

Research Experience: Robot Learning

Dexterous Manipulation from Human Demonstrations — Aug 2026 – Present

Co-advised by Prof. Anqi Wu and Prof. Ye Zhao (LIDAR Lab, Georgia Tech ME)
Platform: 20-DoF Wuji dexterous hands on the Vega humanoid (also Sharpa hands); Newton / MuJoCo Warp and Isaac Lab

- **Goal:** learning contact-rich dexterous skills for real 20-DoF hands from human motion capture, as a skill layer that a high-level VLM or VLA agent can call.
- **Retargeting:** turning human hand–object demonstrations into physically feasible robot references via kinematic retargeting, collision-aware arm IK, and physics-based refinement.
- **Large-scale RL teachers:** privileged-state residual RL policies trained in GPU-parallel simulation, using each demonstration as a prior that RL improves beyond.
- **Sim-to-real:** distilling the privileged RL teachers into a vision-based student policy that runs from camera input on hardware.
- **Hardware validation:** tested the real Wuji hands on the Vega U platform to validate simulator parameters and defined hardware-readiness criteria for deployment.

Research Experience: Learning from Motion and Behavior Data

Motion Encoders and Foundation Models for Animal Behavior — Sep 2025 – Jul 2026

- **ResiFuse** (NeurIPS 2026; first author): A pose-based motion encoder with learned quantization, augmented by video features only on frames with large quantization residuals; outperforms pose- and video-only baselines across three datasets, with the largest gains in multi-agent, socially complex settings.
- **MICE** (NeurIPS 2026 Spotlight): A hierarchical self-supervised foundation model for pose sequences, pretrained jointly on six datasets; a Perceiver-style interaction encoder lets a single checkpoint handle any number of agents.
- **QDMouse4M** (NeurIPS 2026 E&D Track): A six-view, 4M-frame dataset with 2D/3D keypoint labels from subdermal fluorescent markers, for benchmarking pose estimation and behavior analysis.
- **BehaviorVLM** (preprint; co-first author): Finetuning-free pose estimation and behavior understanding by guiding pretrained vision–language models through explicit, verifiable reasoning steps.

Scalable Probabilistic Models of Neural and Behavioral Dynamics Sep 2023 – May 2026

- **ADM** (ICML 2025 **Oral**; first author) and **MRM-GP** (ICML 2024; first author): Gaussian-process state-space models for time-varying, directional, frequency-specific communication between brain regions, with parallel-scan inference reducing time complexity from $O(T^3)$ to $O(\log T)$. Released as **MBRILA**, an open-source GP-SSM framework (MIT license, `github.com/BRAINML-GT/MBRILA`).
- **HDPP** (NeurIPS 2022; first author): Online detection of neural sequences with a hierarchical Dirichlet point process and particle filtering.
- Co-author on **FacRNN** (ICML 2026 Spotlight), **MIG-Vis** (ICLR 2026), **BeNeDiff** (NeurIPS 2024), **POGLM** (ICML 2024), and **VIS** (ICLR 2024 Spotlight): low-rank RNNs, diffusion models for neural latents, differentiable message passing, and variational importance sampling.

Industry Experience

TikTok, Research Intern, Video Recommendation May 2026 – Aug 2026

- Designed and trained large sequence models over multimodal video representations and long user-behavior histories for large-scale recommendation.

Technical Skills

Robotics & Simulation: Isaac Lab / Isaac Sim, Newton / MuJoCo Warp, MuJoCo; kinematic and physics-based motion retargeting; collision-aware IK; residual RL for dexterous manipulation; large-scale RL in GPU-parallel simulation; sim-to-real transfer; multi-fingered hands (Wuji, Sharpa)

Machine Learning: PyTorch; reinforcement and imitation learning; state-space and Gaussian-process models; diffusion models; vision–language models; self-supervised representation learning for motion data

Languages & Tools: Python, C++; AI-assisted development and research workflows with agentic coding tools for experiment design, implementation, and analysis

Publications

*First-author papers marked with †. *Equal contribution.*

† ***Learning When Visual Context Matters for Mouse Behavior Analysis*** **NeurIPS 2026** [link]
Weihan Li, Jingyang Ke, Qinheng Pu, Yule Wang, Chengrui Li, Anqi Wu

MICE: Multi-animal Interaction Context Encoder — A Hierarchical Foundation Model for Mouse Behavior **NeurIPS 2026 Spotlight**
Yen-Shuo Su, ***Weihan Li***, Anqi Wu

QDMouse4M: A Multi-View 3D Mouse Spontaneous Behavior Dataset with Quantum-Dot Markers **NeurIPS 2026 E&D Track**
Jingyang Ke, Amartya Pradhan, Xueling Zhang, ***Weihan Li***, Anqi Wu, Jeffrey E. Markowitz

† ***BehaviorVLM: Unified Finetuning-Free Behavioral Understanding with Vision-Language Reasoning*** preprint [link]
Jingyang Ke*, ***Weihan Li****, Amartya Pradhan, Jeffrey Markowitz, Anqi Wu

A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity **ICML 2026 Spotlight** [link]
Chengrui Li, Yunmiao Wang, Yule Wang, ***Weihan Li***, Dieter Jaeger, Anqi Wu

Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion **ICLR 2026** [link]
Yule Wang, Joseph Yu, Chengrui Li, ***Weihan Li***, Dieter Jaeger, Anqi Wu

† ***Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes*** **ICML 2025 Oral** [link]
Weihan Li, Yule Wang, Chengrui Li, Anqi Wu

Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models **NeurIPS 2024** [link]
Yule Wang, Chengrui Li, ***Weihan Li***, Anqi Wu

† ***Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Communications Across Multiple Brain Regions*** **ICML 2024** [link]
Weihan Li, Chengrui Li, Yule Wang, Anqi Wu

A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing **ICML 2024** [link]
Chengrui Li, ***Weihan Li***, Yule Wang, Anqi Wu

Forward χ^2 Divergence Based Variational Importance Sampling **ICLR 2024 Spotlight** [link]
Chengrui Li, Yule Wang, ***Weihan Li***, Anqi Wu

† ***Online Neural Sequence Detection with Hierarchical Dirichlet Point Process*** **NeurIPS 2022** [link]
Weihan Li, Yu Qi, Gang Pan

† ***Efficient Point-Process Modeling of Spiking Neurons for Neuroprosthesis*** **EMBC 2021** [link]
Weihan Li, Cunle Qian, Yu Qi, Yiwen Wang, Yueming Wang, Gang Pan

Teaching and Service

Teaching: TA, Georgia Tech CSE-8803 Statistical Machine Learning Models for Neural and Behavioral Data Analysis.

Reviewer: NeurIPS 2023–2026, ICML 2024–2026, ICLR 2024–2026.